

IMDB Dataset of 50K Movie Reviews

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Abstract—Convolutional Neural Networks are a type of deep learning models used for feature extraction and detection through convolutional filters. In this work, a working CNN is implemented and trained on a dataset designed for deep learning tasks in order to evaluate its ability to perform classification. Training is analyzed, and the model’s performance is assessed through classification accuracy and learning curves. Additionally, key hyperparameters are tuned, and their effects on model performance are examined and discussed.

Index Terms—CNN, training, Neural Network, analysis

I. INTRODUCTION

CONVOLUTIONAL Neural Networks (CNNs) are a class of feedforward networks that learn features through the optimization of convolutional filters, also known as kernels [1]. CNNs are used in deep learning and have been successful for classification tasks on different types of data including images, text, and audio. The main idea behind CNNs is the extraction of features at different levels of resolution and feature types, where learned feature maps are transformed, flattened, and passed into fully connected layers for the final prediction.

In this assignment, a Convolutional Neural Network is used for sentiment classification using the IMDB movie review dataset. The objective is to predict whether a movie review is a positive or negative sentiment based on text input. CNNs are well-suited for this task, as convolutional filters can capture local patterns and features within word embeddings. The model is implemented using the TensorFlow API and key hyperparameters such as learning rate, kernel size, etc, are adjusted. This report focuses on analyzing how these changes affect the training process and overall model performance and more specifically in which way the change in kernel size may effect the performance results.

A. About Dataset

IMDb is a widely used online database that is an authoritative source for ratings, and user reviews for movies, television shows, and other media content [2]. Its large collection of reviews makes it a useful resource for sentiment analysis and natural language processing tasks.

The IMDB dataset [3] has 50,000 movie reviews labeled for binary sentiment classification. Each sample contains a movie review (text) as the input feature and a corresponding sentiment label indicating whether the review is a positive or negative sentiment. The dataset contains 25,000 highly polar movie reviews for training and 25,000 for testing. Compared

to previous benchmark datasets, this dataset provides substantially more data, making it well-suited for training and evaluating deep learning models.

B. Preprocessing

The dataset is first separated into input features and target variables, where the movie reviews are the input data (X) and the corresponding sentiment labels represent the target variable (y). The review text is then tokenized to convert each review into a numerical vector. The target variable is transformed into 0s and 1s, 0 for negative sentiment and 1 for a positive one. For runtime efficiency, the tokenizer vocabulary is restricted to the top 10,000 most frequent words in the dataset. After, the data is then split into training and testing sets, with 80% of the data used for training and the remaining 20% used for testing.

II. NETWORK ARCHITECTURE

A sequential model is used to initialize the neural network, as layers are stacked linearly. The CNN consists of five layers: an embedding layer, a 1D convolutional layer, a pooling layer, and two fully connected layers. The embedding layer serves as the input stage. The convolutional layer then extracts local features from the embedded layer, which are then reduced in dimensionality by the pooling layer. The final two fully connected layers combine the extracted features and produce the model’s prediction.

Since the input data consist of vectorized text tokens, a 1D convolutional neural network is used to capture local word dependencies within the reviews. The embedding layer maps each word index to a 128-dimensional vector, converting each review into a matrix where rows correspond to word embeddings. During convolution, filters are applied across the matrices to learn local patterns and produce feature maps. Global max pooling is used to reduce each feature map to a fixed-size representation by selecting the maximum activation from each map. These pooled features are then passed through the fully connected layers, where they are integrated to generate the final sentiment classification.

III. TRAINING SETUP: BASELINE MODEL

The input data and corresponding sentiment labels are split into training and testing sets, with the training set containing 40,000 samples. The baseline model is configured with the following params:

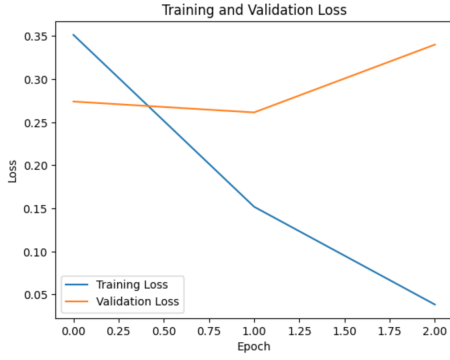


Fig. 1. Baseline Model Loss Curve. Accuracy: 90%

- Embedding dimension: 128
- Convolutional filters: 128
- Dense units: 64
- Learning rate: 0.001

The ReLU activation function is used on the output filters to introduce non-linearity and eliminate any negative values, while the sigmoid function is used in the final classification layer to produce a single output representing the predicted sentiment. Model performance is evaluated using binary cross-entropy loss and classification accuracy. The model is trained for three epochs, with 20% of the training data automatically reserved as a validation set to monitor performance during training and help identify potential overfitting.

IV. EXPERIMENTS & RESULTS

In this section, hyperparameters are modified to evaluate their impact on model performance. Two experiments are conducted to analyze how these changes influence training behavior and accuracy.

A. Baseline Model Performance

After training the model at its baseline configurations, the results were very promising. Achieving an accuracy of 90%, this shows that the model is capable of correctly classifying positive and negative movie reviews in most cases. Further improvements in accuracy could likely be obtained by simply stopping training earlier, as the model already demonstrates strong performance after just a single epoch.

However, the loss curve shown in Figure 1 indicates that the model does not generalize perfectly to unseen data. After the first epoch, the model performs better on training data compared to untrained data. This divergence suggests that the model is learning quickly, which may be due to the initial hyperparameter choices, and could lead to overfitting.

B. Effect of Convolution Kernel Size

In the baseline configuration, the kernel size is set to 5, meaning that a 1D filter spans five tokens at a time. With this configuration, the model obtained an accuracy of 90%. To evaluate the effect of kernel size, both smaller and larger kernel sizes were tested.

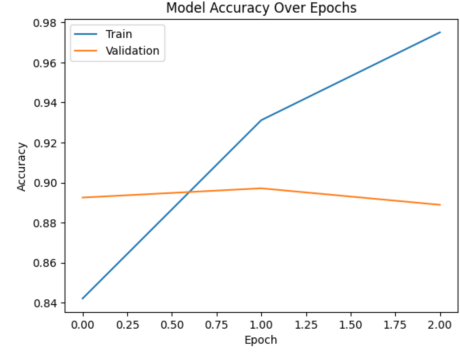


Fig. 2. Model Accuracy Curve for Kernel size: 2

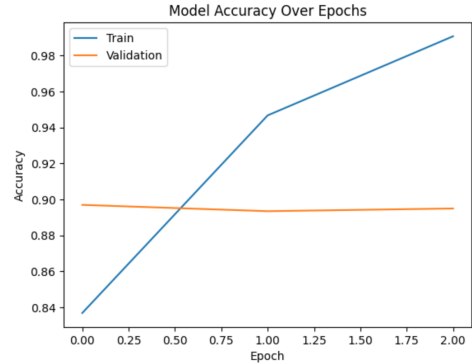


Fig. 3. Model Accuracy Curve for Kernel size: 8

A smaller kernel size of 2 captures extremely local patterns within the text data, focusing on short-range dependencies between adjacent tokens. While such features may be informative, they may not fully capture longer semantic patterns that may be present in the movie reviews. As shown in Figure 2, reducing the kernel size to 2 resulted in an accuracy of 88.8%, which is only slightly lower than the baseline performance.

In contrast, a larger kernel size of 8 allows the model to capture longer-range patterns. However, increasing the kernel size made only minuscule changes in performance. As shown in Figure 3, a kernel size of 8 received an accuracy of 89.4%, which is comparable to the baseline model.

Overall, accuracy remained stable across kernel sizes of 2, 5, and 8, with results consistently falling within the 88-89% range. The primary observed difference was computational time, as larger kernel sizes increased runtime. These results show that kernel size does not have that much of an impact on performance for this dataset, suggesting that local features are sufficient enough for effective classification in the IMDB dataset.

C. Model Complexity Reduction

The baseline model displays a slightly high complexity having over four million parameters. This level of complexity may contribute to overfitting, as the model can also learn noise present in the training data [4]. To address this issue, the following changes were made to the baseline model:

- Embedding dimension: 64

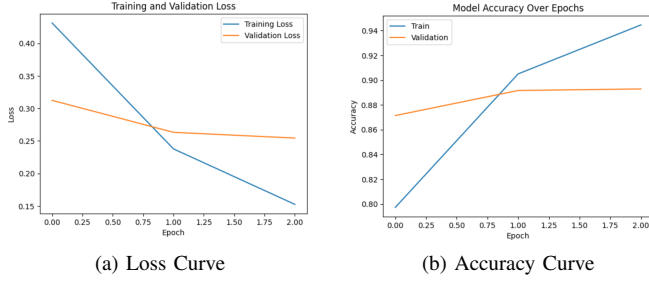


Fig. 4. Results obtained using the Model Configuration.

- Convolutional filters: 64
- Kernel Size: 3
- Dense units: 32
- Learning rate: 0.0005
- 30% Dropout

With these adjustments, the total amount of parameters was reduced to just under two million. Despite the significant reduction in model size, the test accuracy remained high at 89.5% as shown in Figure 4. These result suggests that the parameters that were removed in the simplified model were redundant and did not really contribute to the models performance. Additionally, the reduced model is computationally more efficient than the baseline configuration, making it the preferred architecture for this task.

V. CONCLUSION

This report analyzed the design and performance of a convolutional neural network trained using the IMDB dataset. The baseline model attained strong results after only a single training epoch, indicating that the architecture was able to learn meaningful features from the data. However, later experiments revealed that the baseline model was overparameterized, containing a number of unnecessary parameters that did not contribute much to performance. Reducing the redundant parameters did not have a negative impact on accuracy, which explains why the overall classification performance remained largely unchanged.

Changes to convolutional kernel size also resulted in minimal performance variation. This suggests that smaller more local features dominate for this dataset and increasing the kernel size does not provide substantial additional benefit. As a result, smaller kernels are sufficient enough for capturing the relevant patterns in the text.

Overall, the model demonstrated a strong ability to accurately classify unseen movie reviews, even when using the baseline configuration. Given the large size of the dataset, the model was able to learn quickly. This led to very low training loss and increasing the risk of overfitting with additional epochs. The results indicate that a relatively simple CNN architecture with a single convolutional layer is enough to achieve high accuracy on this task, as meaningful features are extracted early in training, making extended training and increased complexity unnecessary.

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